

$$\ddot{x} + \gamma \dot{x} + x = 0$$

Next step: $\ddot{x} + \gamma \dot{x} + x = f(t)$ ←

→ PSS Q4: ↑ Numerical solⁿ: Computer program

* Energy: Newton: $\vec{F} = m\vec{a} = m \frac{d^2\vec{x}}{dt^2}$
 $\vec{F} = m \frac{d\vec{v}}{dt}$

$$\vec{F} \cdot \vec{v} = \frac{d}{dt}(T)$$

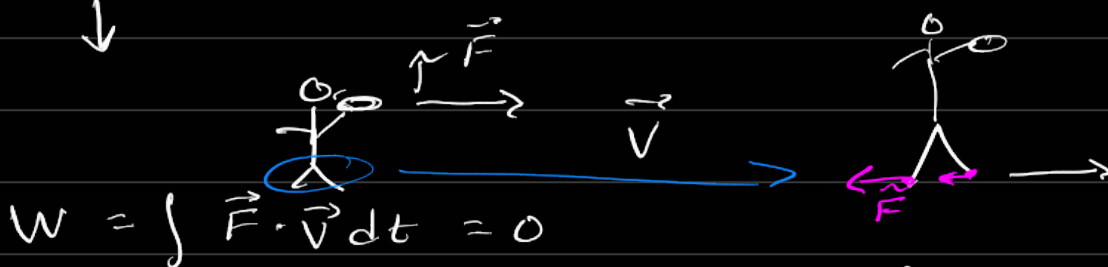
$T =$ kinetic energy
 Energy of Motion

$$T = \frac{1}{2} m v^2$$

$$\int \vec{F} \cdot \vec{v} dt = T$$

Thm: Work done by a force

$$W = \int_{t_1}^{t_2} \vec{F} \cdot \vec{v} dt$$



$$W = \int \vec{F} \cdot \vec{v} dt = 0$$

* Force must cause the motion

$$W = - \int F v dt$$

$$\vec{F} \cdot \vec{v} = |\vec{F}| |\vec{v}| \cos(\pi) = -|\vec{F}| |\vec{v}|$$

Relate to earlier understanding: $W = \int_{t_1}^{t_2} \vec{F} \cdot \vec{v} dt$

$$\int \vec{F} \cdot \frac{d\vec{x}}{dt} dt = \int \vec{F} \cdot d\vec{x}$$

$$d\vec{x} = (dx, dy, dz) = dx \hat{e}_x + dy \hat{e}_y + dz \hat{e}_z$$

two expressions

$$W = \int_{t_1}^{t_2} \vec{F} \cdot \vec{v} dt = \int_{t_1}^{t_2} \vec{F} \cdot \frac{d\vec{x}}{dt} dt = \int_{x_1}^{x_2} \vec{F} \cdot d\vec{x}$$

$W = \int \vec{F} \cdot d\vec{x}$: work has to be in the direction of motion

→ Work ↑

→ K.E.: $T = \frac{1}{2} m v^2$

→ Potential energy: V :

$$F(x) = - \frac{dV(x)}{dx}$$

Back to $W = K.E.$

$$\int \vec{F} \cdot \vec{v} dt = T_2 - T_1 \quad \text{reason}$$

$$\int \vec{F} \cdot d\vec{x} = T_2 - T_1$$

$$\int_{t_1}^{t_2} \frac{d}{dt}(T) dt = T \Big|_{t_1}^{t_2} = T_2 - T_1$$

* Assumption: $\vec{F} \cdot d\vec{x} = F dx$ same direction.

$$\int \mathbf{F} \cdot d\mathbf{x} = T_2 - T_1$$

$$\int \mathbf{F} d\mathbf{x} = \int \left(-\frac{dV}{dx} \right) dx = - \int \frac{dV}{dx} dx = -V \Big|_{x_1}^{x_2}$$

$$= V_1 - V_2$$

$$V_1 - V_2 = T_2 - T_1$$

$$T_1 + V_1 = T_2 + V_2$$

(Conservation of energy)

Initial Energy Final Energy

Call: $T + V = E$ (total Energy)

* Potential Energy: $\int F(x) dx = - \int \frac{dV}{dx} dx$

$$\int F(x) dx = -V$$

* $V = - \int F(x) dx$

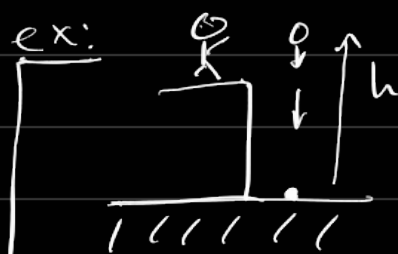
$F_{\text{grav}}: F = -mg$

$$V = \int mg dx = mg \int dx = mgx$$

$$V_{\text{grav}} = mgx$$

$$x = h$$

$$= mgh$$



$$T_i + V_i = T_f + V_f$$

$$\begin{cases} T_i = \frac{1}{2} m v^2 = 0 \\ V_i = mgh \end{cases}$$

$$V_f = mg(0) = 0$$

$$T_f = \frac{1}{2} m v^2$$

$$\bullet mgh = \frac{1}{2} m v^2$$

$$* V = \sqrt{2gh}$$

ex: W/ spring $\square$

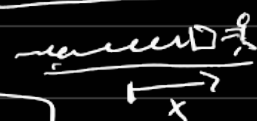
$$F_{\text{Hooke}} = -kx$$

$$W = \int F dx = - \int kx dx = -k \frac{1}{2} x^2$$

$$W = -k \frac{1}{2} x^2$$

$$V = - \int F dx = k \frac{1}{2} x^2$$

x : displacement from equilibrium



(Pg 62) [10-2] $\vec{F}(t) = k \cos(\omega t) \hat{e}_x - mg \hat{e}_y$

(1) Newton's Laws: $\vec{F} = m \frac{d^2 \vec{x}}{dt^2} = m \ddot{x} \hat{e}_x + m \ddot{y} \hat{e}_y + m \ddot{z} \hat{e}_z$

$$\bullet m \ddot{x} = k \cos(\omega t)$$

acceleration

$$m \ddot{y} = -mg$$

$$\ddot{x} = \frac{d}{dt} \dot{x}$$

* Integrate both:

$$\int m \frac{dV_x}{dx} dt = \int k \cos(\omega t) dt$$

$$m(V(t) + C) = \frac{k}{\omega} \sin(\omega t) + \tilde{C}$$

$$V(t)_x = \frac{k}{m\omega} \sin(\omega t) + \underbrace{\left(\frac{\tilde{C}}{m} + C\right)}_{\text{const.}}$$

$V_x \leftarrow \text{constant}$

$$V_x(t) = \frac{k}{m\omega} \sin(\omega t) + V_x$$

• We can also get $x(t)$: $V_x(t) = \frac{dx}{dt}$
one more integral

$$x(t) = \int V_x(t) dt = \frac{k}{m\omega} \int \sin(\omega t) dt + V_x \int dt$$

$$= \frac{k}{m\omega^2} (-\cos(\omega t)) + V_x t + \tilde{C}$$

$$x(t) = -\frac{k}{m\omega^2} \cos(\omega t) + V_x t + x_0$$

Get y and V_y :

$$m\ddot{y} = -mg \Rightarrow \frac{d}{dt}V_y = \ddot{y} = -g$$

$$\text{Integrate: } \int \frac{dV_y}{dt} dt = -g \int dt$$

$$V_y(t) = -gt + V_y$$

find $y(t)$:

$$y(t) = \int V_y(t) dt = -\frac{1}{2}gt^2 + V_y t + y_0$$

All together:

$$\underline{x}$$

$$x(t) = -\frac{k}{m\omega^2} \cos(\omega t) + V_x t + x_0$$

$$V_x(t) = \frac{k}{m\omega} \sin(\omega t) + \underline{V_x}$$

y

$$y(t) = -\frac{1}{2}gt^2 + V_y t + y_0$$

$$V_y(t) = -gt + V_y$$

$$\underline{I.C.}: \vec{r}(0) = \vec{0}$$

$$\underline{\vec{v}(0) = V_x \hat{e}_x}$$

$$\vec{r}(t) = x(t) \hat{e}_x + y(t) \hat{e}_y$$

$$\vec{r}(0) = \vec{0} = x(0) \hat{e}_x + y(0) \hat{e}_y$$

$\uparrow$

$\uparrow$

$$x(0) = 0$$

$$y(0) = 0$$

$$x(0) = -\frac{k}{m\omega^2} + x_0 = 0 \Rightarrow x_0 = \frac{k}{m\omega^2}$$

replace x_0

$$x(t) = -\frac{k}{m\omega^2} (1 - \cos(\omega t)) + V_x t$$

$$x(t) = \frac{m\omega^2}{2} (1 - \cos(\omega t)) + v_x t$$

$$y(0) = y_0 = 0$$

$$y(t) = -\frac{1}{2}gt^2 + v_y t$$

$$\vec{v}(t) = v_x(t) \hat{e}_x + v_y(t) \hat{e}_y$$

$$\vec{v}(0) = v_x \hat{e}_x + 0 \hat{e}_y$$

$$\downarrow$$

$$v_x(0) = v_x$$

$$\downarrow$$

$$v_y(0) = 0$$

$$v_y(0) = -g(0) + v_y = 0 \Rightarrow v_y = 0$$

Finally: y:

$$v_y(t) = -gt$$

$$y(t) = -\frac{1}{2}gt^2$$

